

Team Control Number

office use only

165

Problem Chosen

C

For office use only

F1

F2

F3

F4

2006 Mathematical Contest in Modeling (MCM) Summary Sheet

(Attach a copy of this page to each copy of your solution paper.)

Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

Summary

In this paper, we propose the HIV/AIDS Assessment Model to approximate the expected rate of change in the number of HIV/AIDS infections, and recommend the allocation of financial resources to the United Nations.

Firstly, with no additional interventions taken into account, we divide the people in a special country into three categories: susceptible, infective and exposed. Based on several essential assumptions and the Law of Total Probability, we yield the elements of the transition matrix between the three categories, which reflects their transition characteristics caused by HIV/AIDS. In the same way, we construct another three transition matrixes of natural birth rate, natural death rate and HIV/AIDS mortality. Combining all of the four matrixes leads to the relationship of the category contributions between two adjacent years, which is expected to give reasonable prediction to the HIV/AIDS infections.

Secondly, considering additional interventions (ARV drug therapy, preventative HIV/AIDS vaccine or both) and the development of ARV-resistant disease strains, we regroup the people and adjust the coefficient of the transition matrix.

Then we employ a search algorithm to determine the best allocation of financial resources for the highest rate of susceptible in the country.

The method can be used to the six selected countries respectively. Our model can also be applied to other countries and infections just by modifying the transition matrix; however, the results may fluctuate if the coefficients are not set appropriately.

War of the World: Fight against AIDS

电子科技大学

曾维亮2302001030

梁春生2303304006

彭小帆2310102009

Introduction

There is still a long way to go for nations to increase spending to the levels needed to prevent and treat AIDS. Ninety-five percent of the people with AIDS do not receive medicines that can save their lives.

-Doctor Piot

Over the past two decades, the AIDS epidemic has spread across the world and become an issue so serious that many millions of people are infected, and millions more have died.

The Human Immunodeficiency Virus, which is commonly called HIV, is a virus that directly attacks certain human organs, such as the brain, heart, and kidneys, as well as the human immune system. The immune system is made up of special cells, which are involved in protecting the body from infections and some cancers. The primary cells attacked by HIV can help direct immune function in the body. When enough immune cells have been destroyed by HIV, the immune system barely works. [Brian. 2005]

To fight the HIV/AIDS epidemic, our global society has taken an enormous amount of effort, but there are still some things uncertain about how to approximate the expected rate of change in the number of HIV/AIDS infections and how to most effectively allocate resources to fight this epidemic, which are exactly the aim of this paper.

HIV/AIDS Assessment Model

Four parts are involved in our HIV/AIDS Assessment Model: Country Select Method, Prediction Model, Control Model and Strain Consideration.

Table 1.
Symbol table

PropertySymbolTransition matrix caused by HIV/AIDS A Transition matrix caused by natural birth rate B Transition matrix caused by natural death rate C Transition matrix caused by HIV/AIDS mortality D Total population in year t $N(t)$ the population of Susceptible in year t $S(t)$ the population of Exposed in year t $E(t)$ the population of Infective in year t $I(t)$ the population of Exposed who are cured from Infective in year t $E'(t)$ Nature birth rate in year t b_t Nature death rate in year t d_t every Exposed cause λ person infected annually λ HIV/AIDS mortality in year t e

Country Select Method

With the data provided (income_data.xls, hiv_aids_data.xls and population_data.xls) and Appendix 1, the countries are chosen that are really critical in terms of HIV/AIDS and in dire need of donor assistance.

Select principles are as follows:

- **Countries with high prevalence are prior for purposes of our discussion.** In sub-Saharan Africa, for example, some countries have prevalence even higher than 4%.
- **Countries with low income are especially necessary to be involved in our consideration.** In the low income countries, the overwhelming proportion of HIV-infected persons has no access to treatment, and this has translated into rapidly escalating death rates
- **Only one country is selected for each of the continents (Africa, Asia, Europe, North America, Australia, and South America).**

With the principle above and an extensive search, we make the following countries as our target to study: Ethiopia, India, Albania, Dominican Republic, Fiji and Haiti.

Although the principles may be not suitable for all cases, the specific countries in fact will not affect the models that we develop.

Prediction Model

In order to expect the rate of change in the number of HIV/AIDS infection for these countries from 2006 to 2050, we should know how AIDS occurs. AIDS mainly transmits by blood, such as mother-to-child, prostitution, blood filtering, etc. In our model, we divide it into two way of transmission: mother-to-child transmission and none mother-to-child transmission.

Every year, many people are infected by or die of AIDS. In this model we predict the infection rate of AIDS in the absence of any additional interventions.

Terminology

Latent Period: years when people who are HIV+ are unaware and continue to spread it.

The Susceptible: person not possessing sufficient resistance against a particular pathogenic agent to prevent contracting infection or disease when exposed to the agent, in our model they are health and HIV-.

The Exposed: person who are HIV+, and they are at latent period.

The Infective: person who are HIV+, and they have will die later if no additional interventions are taken.

Assumptions

- **Only the Exposed have the probability to infect Susceptible.** As the Infective has behaved symptom, people will enhance their notice to prevent themselves from infected by Infective.
- **There are no additional interventions:** care interventions & prevention interventions, taken in this model.

Model Formulation

We divide people into three categories (or states): Susceptible (S), Exposed (E), Infective (I). The transition relationship is shown as following:

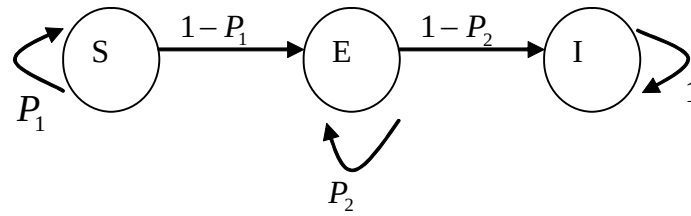


Figure 1. Probability Transition figure.

Assuming that there are $1-P_1$ of Susceptible infected every year, $1-P_2$ of Exposed turn into Infective who will die quickly without treatment. [Stay away from AIDS. 2004]

Transition matrix A can be used to explain the process of epidemic,

$$A = \begin{array}{ccc|c} & S & E & I \\ \hline S & P_1 & 1-P_1 & 0 \\ E & 0 & P_2 & 1-P_2 \\ I & 0 & 0 & 1 \end{array} \begin{array}{l} S \\ E \\ I \end{array}$$

where 0 represents the transition will not happen at all, and 1 represents the transition will occur consequentially. Here

P_1 is not a constant, but effected by the proportion of Exposed in total population. With the increase of $\frac{E(t)}{N(t)}$, there will be more people infected, so we use the following expression to describe this,

$$P_1 = 1 - \lambda \frac{E(t)}{N(t)}, \quad (1)$$

In common, the latent period of AIDS is ten years [Stay away from AIDS. 2004] that means the exposed will turn into the infective before the 10th year on average after they get HIV+. Assuming that the probability of the year that the change occurs follows the uniform distribution, we can yield $1-P_2 = 10\%$.

At the same time, the transition caused by natural birth rate can be implied by transmission matrix B ,

$$B = \begin{bmatrix} 1 & 0 & 0 \\ P_3 & 1-P_3 & 0 \\ P_4 & 1-P_4 & 0 \end{bmatrix}$$

where P_3 represents the probability of infants born by Exposed are Susceptible, then you can explain P_4 in the same way. P_3 & P_4 are effected by the infrastructure of health level in certain country.

And then transmission matrix of the nature death C and HIV/AIDS mortality D are shown as below:

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Note that only the Infective would die of AIDS and all of the people have nature death rate.

Combining all of the four matrixes A, B, C and D leads to the relationship of the category contributions between two adjacent years, which is expected to give reasonable prediction to the HIV/AIDS infections.

$$[S_{t+1}, E_{t+1}, I_{t+1}] = [S_t, E_t, I_t][A + b_t B - d_t C - e_t D] \quad (2)$$

We set Haiti as an example to explain our model. Assuming that in the expression of P_1 , $\lambda = 0.2$ [UNAIDS. 2004], which means every Exposed lead to 0.2 person infected annually. This non-integral value reflects that we are dealing with averages, not individual person.

In Haiti, infection is serious; few pregnant women are offered services for preventing mother-to-child HIV transmission. [UNAIDS. 2004] Here we set $P_3 = P_4 = 0.7$ [Stay away from AIDS. 2004]. In the absence of any additional interventions, the death rate of AIDS will be very high [Stay away from AIDS. 2004], here $e = 90\%$

And from the data of Appendix 1, we find annual nature birth rate b_t and nature death rate d_t . With all the above parameter, we predict the proportion of Susceptible in total population as follows:

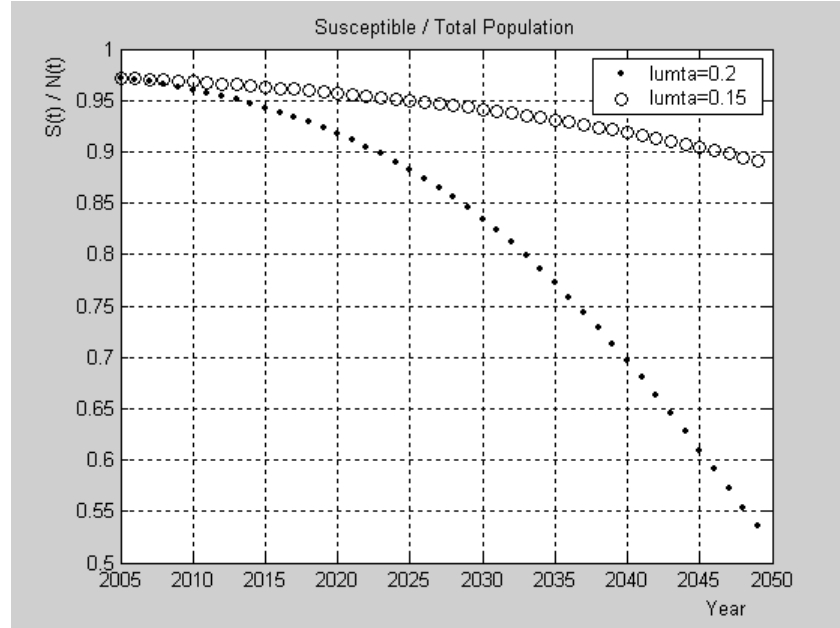


Figure 2: Susceptible/Total Population without any additional interventions.

From this figure, we find that rate of Susceptible to the total population decreases sharply. If we change λ (lumta) from 0.2 to 0.15, there will be 1 in 10 people infected in 2050.

Figure 2 shows that without any additional interventions there will be more and more people infected by HIV/AIDS, and that parameter λ is very sensitivity.

Control Model

In this section we take additional intervention into account. Three parts are involved: Antiretroviral (ARV) Drug Therapy Consideration, Preventative HIV/AIDS Vaccine Consideration, and both of them.

Antiretroviral (ARV) Drug Therapy Consideration

Assumptions

- **International aid is key to financing the AIDS treatment**, which is always more expensive than poor countries can afford. [Adams, Gregor. 2001]
- **Only the Infective receive ARV drug therapy**, the aim of therapy is to prolong the life expectancy of the Infective, but not for the Exposed. [Understanding Epidemics. 2005]
- **The Infective receive DOT (directly observed therapy) for sustaining 3 years, every patient need \$1100 per year.** [Adams, Gregor. 2001]
- **There is no emergence of drug-resistant strains of HIV**, it will be considered later.
- **The Infective who have receive DOT will turn into Exposed and they don't infect other people.** People who have been cured before will be responsible for his act.

Analysis & Model Formulation

The estimate level of financial resources from aid donors is different from country to country, and there maybe arise an alternative case: if HIV/AIDS is controlled before 2050, how will the future funding change? Will it increase with time? It's impossible.

From the reverse aspect, we estimate that the financial funding needed every year is to cure $a\%$ of Infective. From our assumptions, we introduce $E'(t)$, which represents the population of Infective who have received ARV drug

therapy in year t , and they will not impact Susceptible again. A patient who receive ARV drug therapy will last for

3 year, so $\frac{a}{3}\%$ of Infective will turn into Exposed annually on average. The following figure shows the transition relationship vividly,

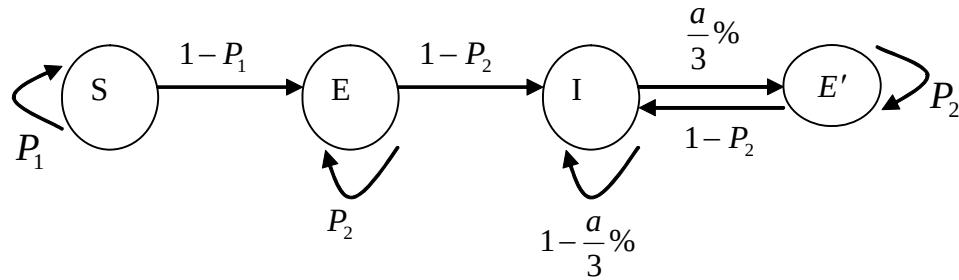


Figure 3. Probability transition

So we fix our matrixes in Prediction Model like this:

$$A = \begin{bmatrix} P_1 & 1-P_1 & 0 & 0 \\ 0 & P_2 & 0 & 1-P_2 \\ 0 & 0 & P_2 & 1-P_2 \\ 0 & 0 & \frac{a}{3}\% & 1-\frac{a}{3}\% \end{bmatrix} \begin{matrix} S \\ E \\ E' \\ I \end{matrix}$$

$$B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ P_3 & 1-P_3 & 0 & 0 \\ P_3 & 1-P_3 & 0 & 0 \\ P_3 & 1-P_3 & 0 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The parameter e will change to e' . As $\frac{a}{3}\%$ of Infective has been cured, $e' = \left(1 - \frac{a}{3}\%\right) \cdot e$ (3)
Other parameters will not change based on our assumptions.

Analogous to the Prediction Model, combining all of the four matrixes leads to the relationship of the category contributions between two adjacent years, which is expected to give reasonable prediction to the HIV/AIDS infections.

$$[S_{t+1}, E_{t+1}, E'_{t+1}, I_{t+1}] = [S_t, E_t, E'_{t+1}, I_t][A + b_t B - d_t C - e_t D] \quad (4)$$

We set Haiti as an example:

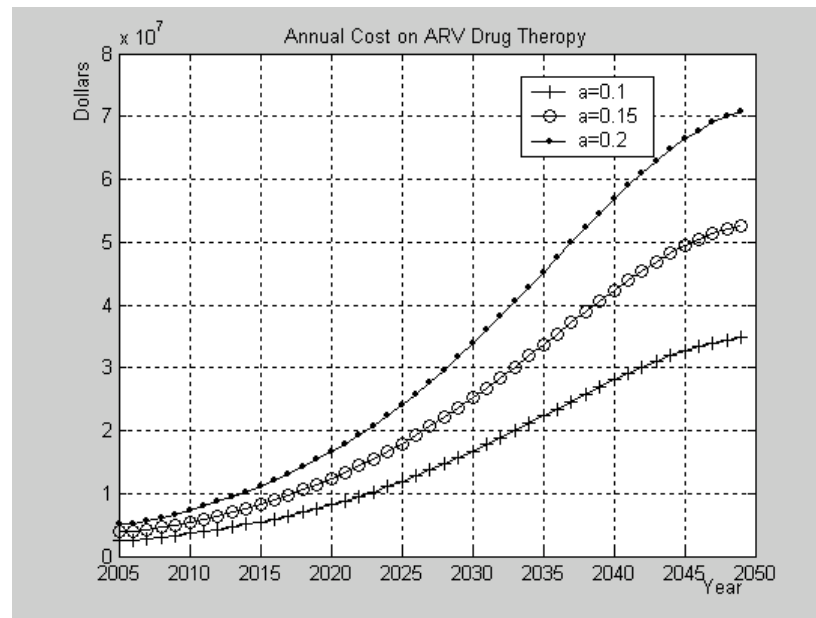


Figure 4. Annual Cost on ARV Drug Therapy with different rate to cure

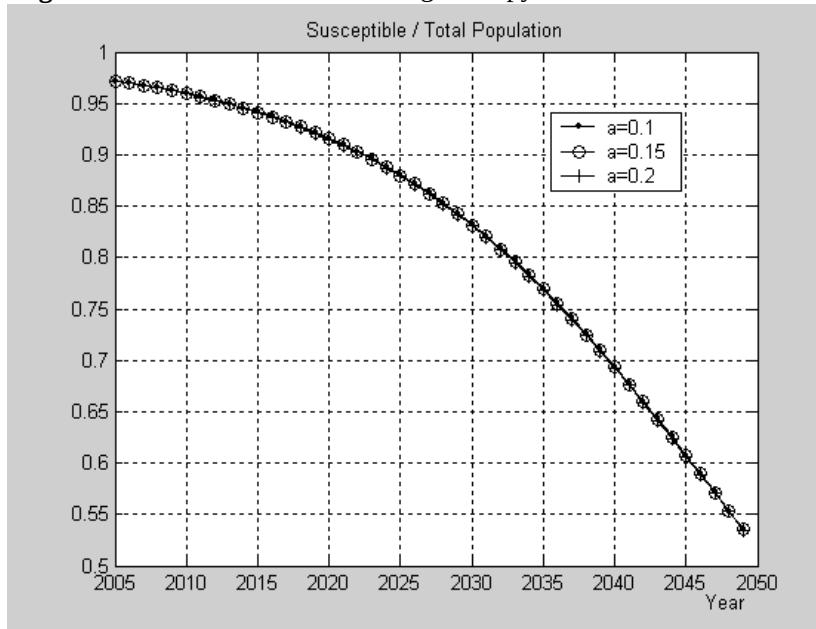


Figure 5. Susceptible/Total Population with ARV Drug Therapy

From susceptible/total population figure, we could not see any difference between different value cure rate a . Even

$$\frac{S(t)}{N(t)}$$

in contrast to Figure 2, where have no intervention, $\frac{S(t)}{N(t)}$ have a little difference. From Figure 4, we must spend more money to cure the more Infective people. The financial resource do not change the proportion of Susceptible in total population, but it is worth to prolong the life of Infective, which protect humane rights and skilled workers for industry development.[2004 report on the global AIDS]

Preventative HIV/AIDS vaccine Consideration

Assumptions

- **HIV/AIDS preventative vaccine will be available in 2010.**
- **The efficacy and duration of protection of the vaccine is 50% and 5 years respectively.** [IAVI. 2005]
- **Vaccination rates might reach the steady-state levels of vaccination in 5 years.**
- **There will be epidemiological externalities from vaccination** that means the externality argument clearly does apply to preventing maternal-child transmission. [Nicholas. 1997]
- **In order to prevent mother-to-child transmission, we vaccinate all the infants,** for the consideration of the

future.

- **The three-dose HIV/AIDS vaccine include the cost of standard package will cost \$20.** [IAVI. 2005]
- **Only the Susceptible can take the vaccine.** In other words there is not effect if the Exposed and Infective take it.

Model Formulation

We assume that $x\%$ of Susceptible have been vaccinated after reaching steady-level of vaccination and the efficacy is 50%. The duration of vaccination is 5 years require that $\frac{x}{5}\%$ of Susceptible should be vaccinated annually to sustain the steady-level. This treatment will change parameter P_1 (Figure 1). Using the Law of Total Probability, we get

$$P_1 = x\% \times 50\% + (1 - x\%) \left(1 - \lambda \frac{E(t)}{N(t)} \right) \quad (5)$$

Note that if $x\% = 0$, the expression above will change back to expression (1).

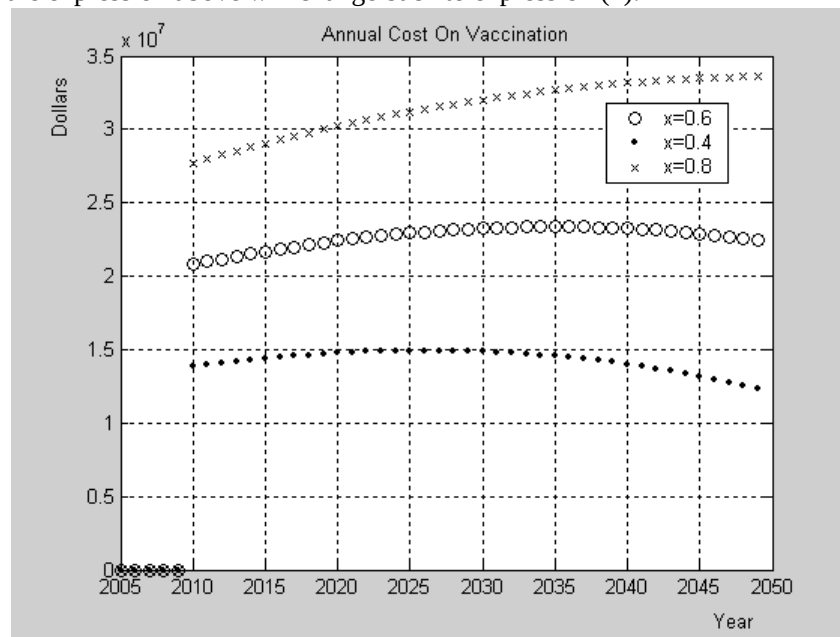


Figure 6. Annual Cost on Vaccination

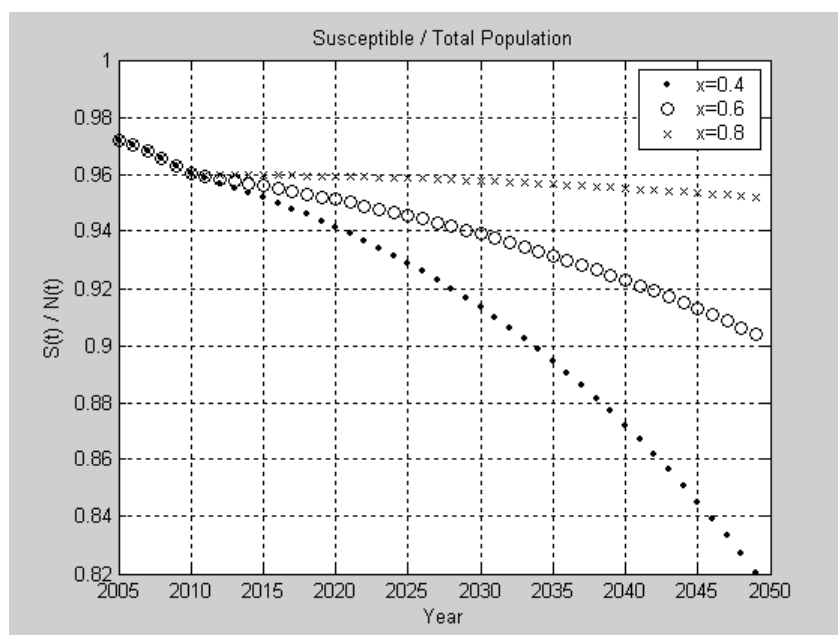


Figure 7. Susceptible/ Total Population with HIV/AIDS Vaccine

The curve changes acutely in year 2010, when HIV/AIDS vaccine is available. When you increase the steady-state levels of vaccination, more money should be spend to immunize the Susceptible, which will reduce the probability of Susceptible changing into Exposed. For example, in order to reach a steady-state 60% vaccination rate, Haiti needs 23 million dollars per year averagely and has 90% of Susceptible in total population in 2050. Although this intervention will not annihilate AIDS, it really staves the transmission of AIDS.

Both consideration

In ARV provision, the parameter e_t (HIV/AIDS mortality in year t) changes; while preventative HIV/AIDS vaccine is taken, P_1 change with time. The two different interventions have different effect on AIDS. ARV drug therapy prolongs the lives of Infective, while HIV/AIDS vaccine decreases the probability of Susceptible changing into Exposed. They act on different population groups and have little relation with each other. Therefore, we can add their effect simply into a matrix. Combined the expression (3) and (5), yielding

$$\begin{cases} e' = \left(1 - \frac{a}{3}\%\right) \cdot 90\% \\ P_1 = x\% \times 50\% + (1 - x\%) \left(1 - \lambda \frac{E(t)}{N(t)}\right) \end{cases}$$

At the same way using the expression (4), we analysis their effect on the proportion of Susceptible in total population:

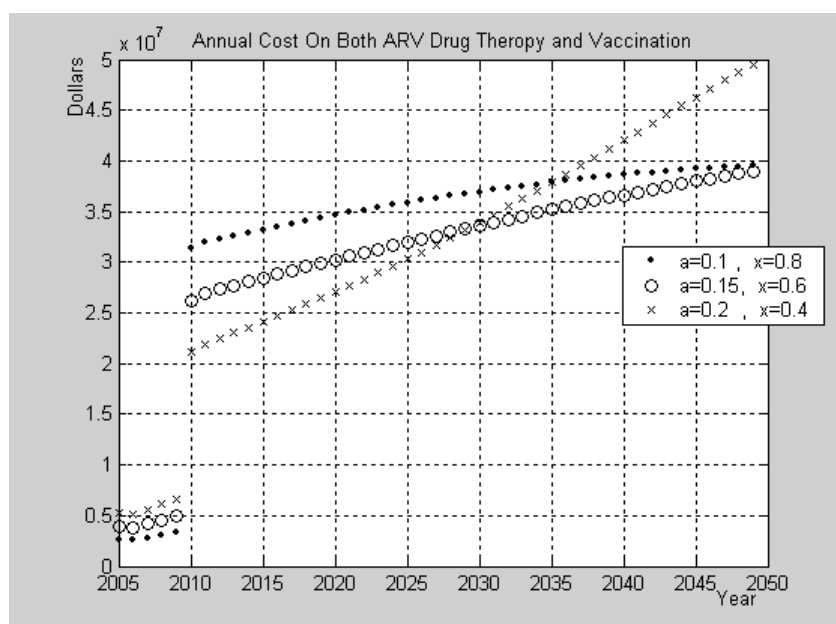


Figure 8. Annual Cost on Both ARV Drug Therapy and Vaccination

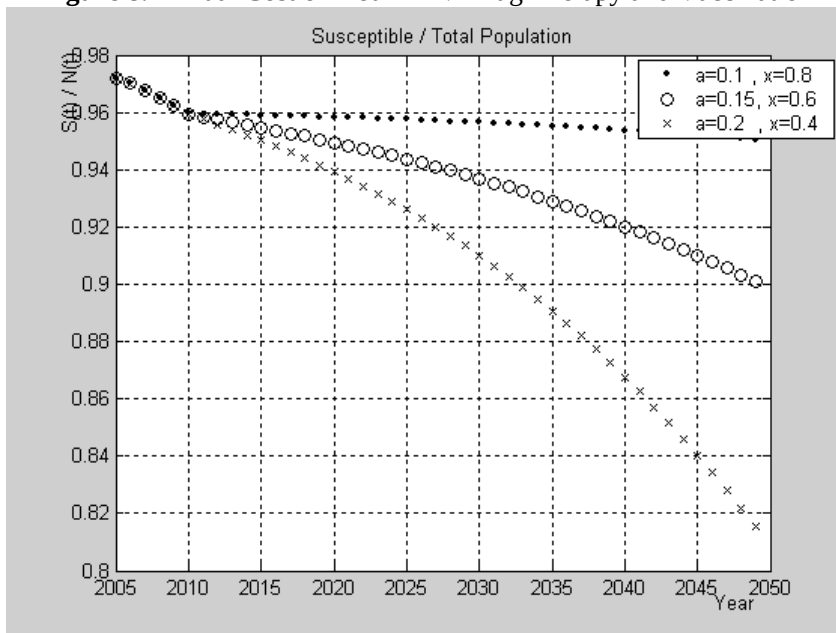


Figure 9. Susceptible/Total Population with Both ARV Drug Therapy and Vaccination

Compared with the above alternative interventions, the use of both ARV provision and a preventative HIV/AIDS vaccine will get a fantasy effect to prolong the longevity of Infective and prevent the transmission of AIDS. As of the financial restrains, a steady-state vaccination of 60% and cure rate of 15% maybe a better choice.

Strain Consideration

Terminology on ARV-resistant Disease Strains

First-line treatment: the preferred therapy for a particular condition, in other word it means the therapy offered initially following diagnosis. [Amfar AIDS research]

Second-line treatment: treatment that is given after the disease has not responded to a first course of therapy or a patient ceases first-line of therapy.

Strains Analysis

ARV drug therapies have tremendous benefits in terms of prolonging the lives of individuals infected with HIV/AIDS. However, the efficacy of ARV depends on the adherence to the treatment regimen and to proper monitoring. Non-adherence or inadequate treatment leads to two very serious consequences. First, the treatment may not be effective for the individual received treatment. Second, partial or inadequate treatments are thought to directly

lead to the emergence of drug-resistant strains of HIV. We assume that a person receiving ARV treatment with adherence below 90 percent has a 5 percent chance of producing a strain of HIV/AIDS.

In order to explain the importance of adherence to treatment regimen and effective monitoring, we estimate the proportional rate of AIDS death to total population.

Considering there is $a\%$ of Susceptible who receive ARV treatment, $y\%$ of them with adherence below 90 percent.

Antiretroviral (ARV) drug therapy

If there is no drug-resistant strains of HIV, ARV drug therapy will make $\frac{a}{3}\%$ of Infective turn into Exposed annually. When there are drug-resistant strains, only the Infective who receive ARV treatment with adherence above 90 percent will turn into Exposed. So the matrix A of Control Model will be changed like this:

$$A = \begin{bmatrix} P_1 & 1-P_1 & 0 & 0 \\ 0 & P_2 & 0 & 1-P_2 \\ 0 & 0 & P_2 & 1-P_2 \\ 0 & 0 & \frac{a}{3}\%(1-y\%)(1-5\%) & 1-\frac{a}{3}\%(1-y\%)(1-5\%) \end{bmatrix}$$

At the same time,

$$e' = \left(1 - \frac{a}{3}\%(1-y\%)(1-5\%)\right) \cdot 90\% \quad (6)$$

And the matrixes B, C and D will not change.

Using the expression (4), we analysis what the drug-resistant strains will effect death rate which cause by AIDS.

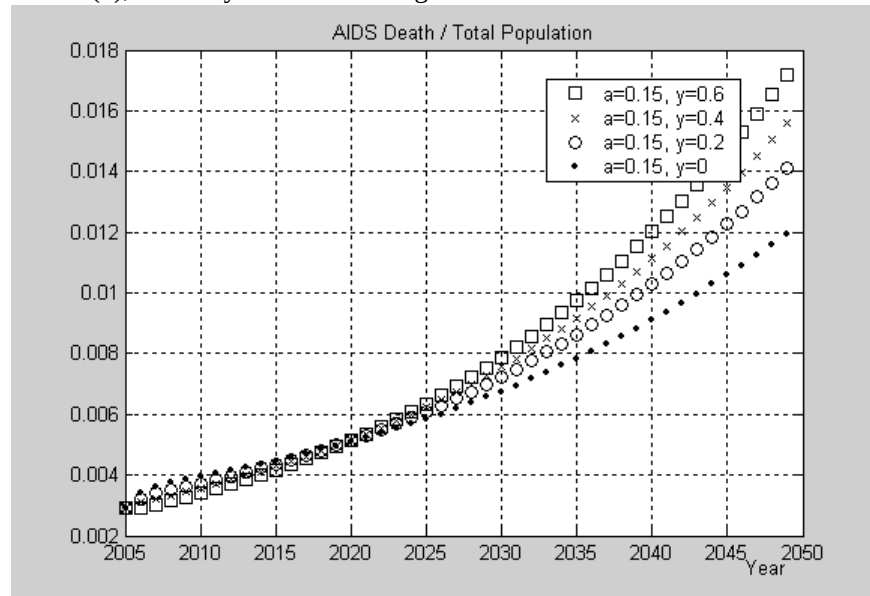


Figure 10. AIDS Death/Total Population with Drug Strain Caused by Antiretroviral (ARV) drug therapy

With a steady-state cure rate of 15% per year, these graphs show that the ratio of AIDS death to total people follows the change of drug-resistant strains of HIV to a certain extent. Hence, avoiding the interruption in the therapy is essential to insure the effort of therapy and save cost.

Preventative HIV/AIDS vaccine

There is no Antiretroviral (ARV) drug therapy, so we assume there are no drug-resistant strains.

Both ARV provision and a preventative HIV/AIDS vaccine

As we have analysis in Control Model, combined the expression (5) and (6) yield,

$$\begin{cases} e' = \left(1 - \frac{a}{3}\% \cdot (1 - y\%)(1 - 5\%)\right) \cdot 90\% \\ P_1 = x\% \times 50\% + (1 - x\%) \left(1 - \lambda \frac{E(t)}{N(t)}\right) \end{cases}$$

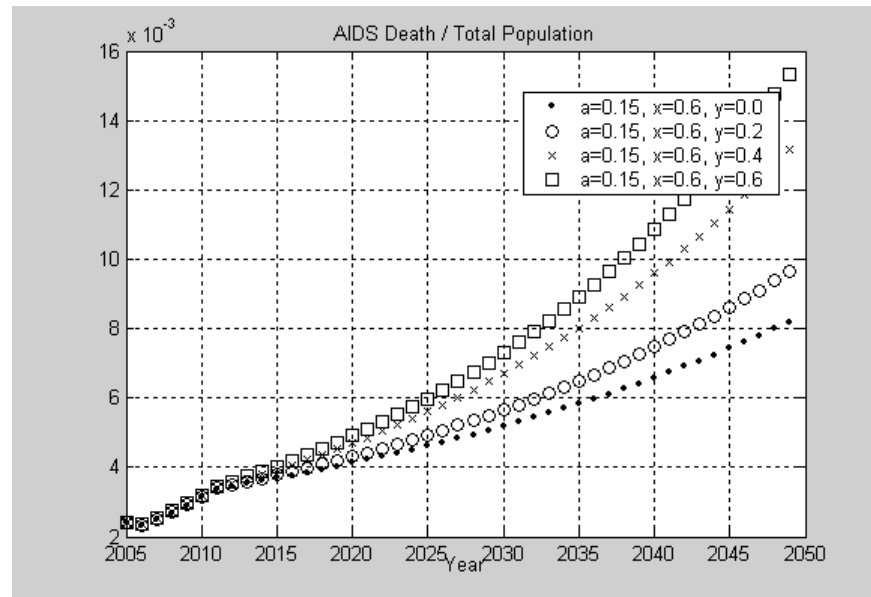


Figure 11. AIDS Death/Total Population with Both ARV Drug Therapy and Vaccination

When both the ARV drug therapy and preventative HIV/AIDS vaccine Consideration are available, supposing that a steady-state vaccination of 60% and cure rate of 15%, the ratio of AIDS death to total people also increases with the higher drug-resistant strains. However, the ratio increases not so rapidly, comparing to the condition of only the ARV drug therapy measure. Clearly, it is caused by the prevention from HIV/AIDS vaccine.

Method for Allocation of Financial Resources

If there are X billion dollars from international aid for the six special countries, the following method for allocation is given in order to take fully advantage of the financial resources from donor.

Firstly, we apply Analytic Hierarchy Process to determine the weight of the allocation money for every special country. The income of the country and the prevalence rate are taken into account. From this we can yield how much each country can get from UN.

Secondly, for every country we employ an extensive searching arithmetic to determine how much to spend on Antiretroviral (ARV) Drug Therapy and Preventative HIV/AIDS vaccine respectively so that the proportion of the Susceptible in the country can be highest.

Strengths and Weaknesses of the Model

Strengths

- With a little data, and some mathematics, an excellent predictive model can be made. Our model should be a useful tool in decision making on HIV/AIDS.
- Our model can be widely used for all kinds of infections just by fixing the coefficient of the matrix, which is determined by our knowledge of the infections.

Weaknesses

- The prediction is sensitive to the coefficient of our model in that reliable data on HIV prevalence rates by county are generally difficult to obtain. But before you make a prediction with our model, you must do an extensive research to determine the exact value.

- In the transition matrix P_1 is estimated that it has a linear relationship to $\frac{E(t)}{N(t)}$ which is maybe not exact.

White Paper for UN

The Importance of HIV/AIDS

We argue that there are essentially four primary reasons why the whole world should be interested and involved in the struggle against AIDS

- **HIV/AIDS is a development issue, not just a health issue.** HIV/AIDS affects not just the health status, but also the social, economic and psychological well being of individuals and society. It has the capacity to weaken economies by diminishing the supply of educated, skilled and professionally qualified people and reducing the pool of leaders in critical areas of society, the economy and government. [UK Department for International Development. 2001]
- **HIV/AIDS affects not only individuals but also systems and the whole world.** Twenty-five years into the HIV/AIDS epidemic, its impact on the populations of the highly-affected countries is becoming increasingly evident. Despite an enormous amount of effort, our global society remains uncertain on how to control the epidemic effectively.
- **HIV/AIDS affects human resource development.** Every year a lot of people die, but are far away from reaching their natural life expectancies, lose their lives in AIDS.
- **The struggle against HIV/AIDS requires International Brainstorming.** It is not the responsibility of any special country, but the entire world. Preventing the spread of HIV/AIDS, managing its impacts and supporting those infected with HIV/AIDS require constant attention to research and development.

Allocation of the Resources

About 90 percent of infections occur in developing countries, where the disease has already reduced life expectancy, in some cases by more than a decade. HIV is already widespread in many countries in Sub-Saharan Africa and may be on the verge of exploding in other regions. Because most people who develop AIDS are adults in the prime of life, the disease exacts a heavy toll on surviving family members, especially children, and may exacerbate poverty and inequality. Clearly, the human toll of the epidemic is great. But low-income countries face a multitude of pressing human needs. [Peter, James. 2002] How should the international community especially the United Nations (UN) respond?

Based on our model and analysis, we make the following recommends to UN for the resource allocation on three parts: Prevention, Treatment and Research.

- **Efficient prevention is critical to stop the infection of ADIS.** Effective prevention before HIV/AIDS becomes evident can avert suffering, save lives, and avoid potentially massive expenditures on AIDS treatment and care. Each case of HIV infection directly prevented among people who practice these high-risk behaviors will indirectly prevent many secondary infections in the rest of the population-a kind of "multiplier" effect. Others in the population who practice lower-risk behavior by having few partners, consistently using condoms, or using sterilized injection equipment are unlikely to spread HIV, even if they contract HIV themselves. The likelihood that an individual will contract and spread HIV is determined by the level of the individual's risk behavior. Behavioral studies show that observable individual characteristics, such as occupation, age, or sexual orientation, can partially predict risk behavior and therefore can be useful in guiding prevention efforts. However, those with the riskiest behavior vary from country to country and over time.
- **Treatment is necessary to save the labor and protect the children.** If AIDS kills lots of the labor, the productivity of country will be suffered drastically. So, the treatment is necessary to prolong the people's lifespan who has been infected. In this way, they can devote their abilities to society. In the other hand, it also can protect their children from the potential threat.
- **Quicken the research and development of the AIDS vaccine.** As we known, the prevention makes a good effort for struggle against the AIDS. If a preventative HIV vaccine is available, we can inject the newborn baby to prevent the AIDS fundamentally, such as diphtheria, tetanus and pertussis.

Coordinate Donor Involvement for HIV/AIDS

Important progress has been made in raising additional funds to respond to the AIDS epidemic. By 2003, an estimated US\$ 5 billion was available, from donors, the UN system, international nongovernmental organizations, country governments and the 'out-of-pocket' spending by people living with HIV and their families. Yet this amount is less than half of what is required by 2005. [UNAIDS. 2004]

Two-thirds of global funding for 2005 and subsequent years is expected to come from the international community. Most of this money will be spent to meet the needs of the poorest and worst-affected countries of Asia and sub-Saharan Africa; these countries will rely on external donors to meet up to 80% of their needs.

Vaccines and microbicides are global public goods (goods that benefit others beyond those who use them directly); each prevented infection cuts off a potential chain of infections resulting from the primary infection. Both private and public sector investment is needed for vaccines and microbicides. According to the International AIDS Vaccine Initiative, public sector investment in vaccine research looks set to expand but overall funding is not keeping up with

the challenges.

We believe that on moral, health, social and economic grounds the international community should provide the scientific and financial leadership for a rapid scaling-up of AIDS treatment in the poorest and hardest-hit countries of the world.

Conclusions

The estimate to rate of change in the number of HIV/AIDS infections is particular important to describe the AIDS epidemic of the whole world. The method discussed in this paper is based on the analysis of the relationship between two adjacent years. We design a model that is capable and adaptable to predict any other epidemic of the world.

References

Adams, Gregor. 2001. Consensus Statement on Antiretroviral Treatment for AIDS in Poor Countries.

Amfar AIDS research. <http://www.amfar.org/cgi-bin/iowa/bridge.html> <<http://www.google.com/url?sa=X&start=0&oi=define&q=http://www.amfar.org/cgi-bin/iowa/bridge.html%3Fpage%3DF?&page=F>>. Accessed on 5 February 2006

Brian. 2005. Introduction to HIV: What are HIV and AIDS? http://llcare.org/focus_article.asp?f=hiv&b=healthology&c=hiv_intro. Accessed on 5 February 2006

IAVI. 2005. Advance Market Commitments: Helping to accelerate AIDS vaccine development

Nicholas. 1997. Economist Setting Priorities for Government Involvement with Antiretroviral

V. G. Kulkarni. 2002. OPERATIONS RESEARCH: Applied Stochastic Models. Springer

Peter, James. 2002. Strategic Lessons from the Epidemiology of HIV
<http://www.worldbank.org/aids-econ/confront/confrontfull/chapter2.html>. Accessed on 7 February 2006

SCIENCE IN THE NEWS - A New Plan to fight AIDS. 2003.
<http://www.unsv.com/voanews/specialenglish/scripts/2003/06/10/0045/>. Accessed on 5 February 2006

Stay away from AIDS. 2004. <<http://www.hangzhou.com.cn/20040101/ca513081.htm>>. Accessed on 6 February 2006

UK Department for International Development. 2001. Commonwealth Universities in the Age of HIV/AIDS: What Every Senior Executive Needs to Know. <http://www.ncsu.edu/ncsu/aern/comonka.html>. Accessed on 6 February 2006.

UNAIDS. 2004. Report on the global epidemic AIDS

Understanding Epidemics - Section 2: HIV/AIDS - Control. 2005.
<http://www.liv.ac.uk/geography/research_projects/epidemics/HIV_control.html>. Accessed on 7 February 2006

United Nations. 2004. World Population Prospects: The 2004 Revision.

Appendix 1: Data Table used

Table 2

Total population (both sexes combined) by country, annually for 1999-2050 (thousands)

Country	1999	2000	2001	...	2049	2050
Albania	3063.18	33061.77	53066.36	...	3467.18	93458.10
Dominican Republic	8143.36	28264.91	28388.46	...	12644.71	12668.10
Ethiopia	66817.62	68525.12	70258.95	...	168225.55	170190.41
Fiji	802.62	3810.73	6818.65	...	938.02	3934.41
Haiti	7825.58	7938.79	18053.27	...	12933.21	12996.03
India	1004199.81	1021084.21	1037808.6	...	1588596.62	1592704.13

Table 3

Crude birth rate by country, 1999-2050 (births per thousand population)

Country	1995-2000	2000-2005	2005-2010	...	2040-2045	2045-2050
Ethiopia	43.69	41.13	363.93	...	22.43	820.89
India	27.17	24.57	322.52	...	13.07	112.39
Albania	18.86	17.22	516.91	...	11.75	711.24
Haiti	31.62	730.44	829.06	...	16.77	15.62
Dominican Republic	25.74	824.46	422.94	...	13.49	412.81
Fiji	25.84	823.55	821.62	...	12.99	612.13

Table 4
The estimated number of HIV positive 0 to 49 year olds at the end of 1999

country#	HIV+, 1999	Dominican Republic	130000	Ethiopia	3000000	Fiji	300	Haiti	210000	India	3700000	Albania	100
----------	------------	--------------------	--------	----------	---------	------	-----	-------	--------	-------	---------	---------	-----

countrythird diptheria-tetanus-pertussis dose vaccination coverage rate, 2002second tetanus toxoid "plus" vaccination coverage rate, 2002Dominican Republic81.686Ethiopia4924Fiji9298Haiti38.6525.6India7086Albania9871**Table 6**
Gross national product (GNP)

countryWorld Bank income codegross national product (GNP) per capita, 2002Dominican Republic22130Ethiopia1100Fiji21820Haiti1510India1450Albania11120

Appendix 2: Some Matlab programs for our solution

Initialization Data

```
% InitData.m
% The matrix format:
% Row: Country: Dominican      Ethiopia      Fiji      Haiti      India      Albania
% Column: Year: 1999 - 2050
% Some Detail omitted for the vast matrix
Population = [...
8143.362      8264.912      8388.462 ...
66817.62      68525.124      70258.953 ...
802.623 810.736 818.656 ...
7825.58 7938.791      8053.271 ...
1004199.868 1021084.243 1037808.693 ...
3063.183      3061.775      3066.367 ...];
Population = Population * 1000; % Because the original data's units is thousand.
BirthRate1 = [...];
BirthRate1 = BirthRate1 * 0.001; % Because the original data's units is per thousand population.
% Calculate the population increment Rate
for country = 1 : 6,
    for year = 1 : 52 - 1,
        PopuRate(country, year) = Population(country, year + 1) / Population(country, year);
    end
end
PopuRate(:, 52) = PopuRate(:, 52 - 1); % The last item is not available
for country = 1 : 6,
    for stage= 1 : length(BirthRate1),
        if stage== 1,
            BirthRate(country, 1) = BirthRate1(country, 1);
            BirthRate(country, 2) = BirthRate1(country, 1);
        else for year = 1 : 5,
            BirthRate(country, 2 + 5 * (stage- 2) + year ) = BirthRate1(country, stange);
        end
    end
end
end
% Calculate the Nature Death Rate
DeathRate = ones(6, 52) + BirthRate - PopuRate; % The last item is not available
S = zeros(6,52);
E0 = zeros(6,52);E1 = zeros(6,52);
I = zeros(6,52);
E1(:,7) = [ 0.1530;3.4168;0.0003;0.2388;4.4236;0.0001].*(10^6) .* 0.9;
I(:,7) = [ 0.1530;3.4168;0.0003;0.2388;4.4236;0.0001].*(10^6) .* 0.1 ;
S(:,7) = Population(:,7) - E0(:,7) - E1(:,7) - I(:,7);
N = zeros(6,52);
rumta = zeros(6,52);
Money = zeros(6,52);
```

Predict the susceptible, exposed and infective from 1999 to 2050 only with Antiretroviral (ARV) drug therapy

```
% ARV_Only.m
Clear; InitData;
```

```

% init. the S, E, I.

% New Variable
DeadPeople = zeros(6,52);
% New Constant
a = 0.12;
for country = 1 : 6,
    for year = 1 : 52 - 1,

        AnnualMoney = 1100;
        if year < 7,
            continue
        end

        N(country,year) = ...
            sum([S(country, year),E1(country,year), E0(country, year),I(country, year)]);
        rumta(country,year) = 0.15;

        P1 = 1 - rumta(country,year) * (E1(country, year) / N(country,year));

        P2 = 0.9;
        P3 = 0.7;
        P4 = P3;
        b = BirthRate(country, year);
        d = DeathRate(country, year);
        e = 0.8;
        e = (1 - a / 3.0) * e;

        % Disease
        % S    E1    E0    I
        A = [P1, 1 - P1,    0,    0;
            0, P2,    0,    1 - P2;
            0, 0,    P2,    1 - P2;
            0, 0,    a / 3.0, 1 - a / 3.0];

        % Nature Birth
        B = [1, 0,    0, 0;
            P3, 1 - P3, 0, 0;
            P3, 1 - P3, 0, 0;
            P4, 1 - P4, 0, 0];

        % Nature Death
        C = [1, 0, 0, 0;
            0, 1, 0, 0;
            0, 0, 1, 0;
            0, 0, 0, 1];

        % Infected
        D = [0, 0, 0, 0;
            0, 0, 0, 0;
            0, 0, 0, 0;
            0, 0, 0, 1];

        TransMat = A + b * B - d * C - e * D;
        Money(country, year) = a * I(country, year) * AnnualMoney;
        DeadPeople(country, year) = I(country,year) * e;
        SEI = ...
            [S(country, year),E1(country, year),E0(country, year),I(country, year)] * TransMat;
        S(country, year + 1) = SEI(1);
    end
end

```

```

    E1(country, year + 1) = SEI(2);
    E0(country, year + 1) = SEI(3);
    I(country, year + 1) = SEI(4);
end % end year
end % end country
N(:,end) = N(:,end - 1);
rumta(:,end) = rumta(:,end - 1);
Money(:, end) = Money(:,end - 1);
DeadPeople(:,end) = DeadPeople(:,end - 1);
disp('Complete!');
for country = 1 : 6,
    % Print the essential data and graph.
    figure(1);plot(2005:2050 - 1, S(country,7:end - 1));title('Susceptible');grid on;
    ...
    % Detail Omitted
end

```

Predict the susceptible, exposed and infective from 1999 to 2050 only with a preventative HIV/AIDS vaccine available in 2010

```

% Vaccine _Only.m
Clear; InitData;
% init. the S, E, I.
% New Variable
DeadPeople = zeros(6,52);
% New Constant
a = 0.12;
X = 0.5;
AnnualMoney = 1100;
for country = 1 : 6,
    %disp("");
    for year = 1 : 52 - 1,

        if year < 7,
            continue
        end
        if year < 12
            x = 0;
        else x = X;
        end
        N(country,year) = ...
            sum([S(country, year),E1(country,year), E0(country, year),I(country, year)]);
        rumta(country,year) = 0.2;

        P1 = 1 - rumta(country,year) * (E1(country, year) / N(country,year));
        P1 = x / 2 + (1 - x / 2) * P1;

        P2 = 0.9;
        P3 = 0.9;
        P4 = P3;
        b = BirthRate(country, year);
        d = DeathRate(country, year);
        e = 0.8;
        e = (1 - a / 3.0) * e;

        % Disease
        % S    E1    E0    I
        A = [P1, 1 - P1, 0,    0;
            0, P2,    0,    1 - P2;

```



```

    0, 0,    P2,    1 - P2;
    0, 0,    a / 3.0, 1 - a / 3.0];

% Nature Birth
B = [1, 0,    0, 0;
     P3, 1 - P3, 0, 0;
     P3, 1 - P3, 0, 0;
     P4, 1 - P4, 0, 0];

% Nature Death
C = [1, 0, 0, 0;
     0, 1, 0, 0;
     0, 0, 1, 0;
     0, 0, 0, 1];

% Infected
D = [0, 0, 0, 0;
     0, 0, 0, 0;
     0, 0, 0, 0;
     0, 0, 0, 1];

TransMat = A + b * B - d * C - e * D;
Money(country, year) = (x/5)*S(country,year) * 20;
DeadPeople(country, year) = I(country,year) * e;
SEI = [S(country, year),E1(country, year),E0(country, year),I(country, year)] * TransMat;
S(country, year + 1) = SEI(1);
E1(country, year + 1) = SEI(2);
E0(country, year + 1) = SEI(3);
I(country, year + 1) = SEI(4);
end % end year
end % end country
N(:,end) = N(:,end - 1);
rumta(:,end) = rumta(:,end - 1);
Money(:, end) = Money(:,end - 1);
DeadPeople(:,end) = DeadPeople(:,end - 1);
disp('Complete!');
for country = 1 : 6,
    % Print the essential data and graph.
    figure(1);plot(2005:2050 - 1, S(country,7:end - 1));title('Susceptible');grid on;
    ...
    % Detail Omitted
end

```

Predict the susceptible, exposed and infective from 1999 to 2050 with both Antiretroviral (ARV) drug therapy and a preventative HIV/AIDS vaccine available in 2010

```

% ARV_Vaccine .m
Clear; InitData;
% init. the S, E, I.
a = 0.1;
X = 0.5;
AnnualMoney = 1100;
for country = 1 : 6,
    %disp("");
    for year = 1 : 52 - 1,
        if year < 7,
            continue
        end
        if year < 12
            x = 0;

```

```

else x = X;
N(country,year) = ...
    sum([S(country, year),E1(country,year), E0(country, year),I(country, year)]);
rumta(country,year) = 0.2;

P1 = 1 - rumta(country,year) * (E1(country, year) / N(country,year));
P1 = x / 2 + (1 - x / 2) * P1;

P2 = 0.9;
P3 = 0.9;
P4 = P3;
b = BirthRate(country, year);
d = DeathRate(country, year);
e = 0.8;
e = (1 - a / 3.0) * e;

% Disease
% S    E1    E0    I
A = [P1, 1 - P1, 0,    0;
     0, P2,    0,    1 - P2;
     0, 0,    P2,    1 - P2;
     0, 0,    a / 3.0, 1 - a / 3.0];

% Nature Birth
B = [1, 0,    0, 0;
     P3, 1 - P3, 0, 0;
     P3, 1 - P3, 0, 0;
     P4, 1 - P4, 0, 0];

% Nature Death
C = [1, 0, 0, 0;
     0, 1, 0, 0;
     0, 0, 1, 0;
     0, 0, 0, 1];

% Infected
D = [0, 0, 0, 0;
     0, 0, 0, 0;
     0, 0, 0, 0;
     0, 0, 0, 1];

TransMat = A + b * B - d * C - e * D;
Money(country, year) = (x/5)*S(country,year) * 20 + a * I(country,year) * AnnualMoney;
DeadPeople(country, year) = I(country,year) * e;
SEI = [S(country, year),E1(country, year),E0(country, year),I(country, year)] * TransMat;
S(country, year + 1) = SEI(1);
E1(country, year + 1) = SEI(2);
E0(country, year + 1) = SEI(3);
I(country, year + 1) = SEI(4);
end % end year
end % end country
N(:,end) = N(:,end - 1);
rumta(:,end) = rumta(:,end - 1);
Money(:, end) = Money(:,end - 1);
DeadPeople(:,end) = DeadPeople(:,end - 1);
disp('Complete!');
for country = 1 : 6,
    % Print the essential data and graph.
    figure(1);plot(2005:2050 - 1, S(country,7:end - 1));title('Susceptible');grid on;

```

```
... % Detail Omitted
end
```